

Sourav Yadav

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CAREER OBJECTIVE

Software Developer with hands-on experience building full-stack applications across the MERN and Java/Jakarta EE stacks, and deploying them with modern DevOps practices including Docker, AWS EC2, and CI/CD pipelines through GitHub Actions. Comfortable working across the stack, from relational and NoSQL database design to REST API development and containerized deployment. Driven by clean, maintainable code and a continuous interest in emerging technologies, with a track record of turning project ideas into working, production-style systems.

EDUCATION

BSc (Hons) Computing <i>Islington College London Metropolitan University, Kathmandu</i>	2024 - Present
SLC <i>Golden Gate International Secondary School NEB, Kathmandu</i>	2020 - 2022
SEE <i>Loyalty Academy NEB, Budhanilkantha, Kathmandu</i>	2022

SKILLS

Languages: Java, Python, JavaScript, SQL
Frontend: React, HTML5, CSS, Tailwind CSS
Backend: Node.js, Express.js, Java (Jakarta EE)
Database: MongoDB, MySQL, Oracle
DevOps & Cloud: Docker, Docker Compose, AWS EC2, Nginx, GitHub Actions (CI/CD)
Libraries & Tools: Socket.io, JWT, Bcrypt, Axios, JSTL, JDBC, VS Code, Eclipse, IntelliJ, Figma
Data & Analysis: NumPy, Pandas, Matplotlib, Seaborn, SciPy, Statistical Analysis, Feature Engineering
Version Control: Git, GitHub
Soft Skills: Teamwork, Problem Solving, Critical Thinking, Communication, Adaptability, Analytical Thinking

PROJECTS

Personal Projects

<u>StudyMate — AI-Powered MERN Study Companion</u>	2026
Language and Tech Used: <i>MongoDB, Express.js, React, Node.js, Socket.io, JWT, Google Gemini AI</i>	
- Built a full-stack MERN application that lets users upload study documents (PDF, Word, PPT) and auto-generates flashcards and quizzes from them using the Google Gemini AI API. - Implemented real-time multiplayer quiz rooms with Socket.io, JWT-based authentication, and topic-mastery tracking with progress analytics dashboards built using Recharts. - Designed RESTful APIs and MongoDB schemas (Mongoose) covering documents, quizzes, flashcards, study rooms, and study logs.	

Language and Tech Used: *Node.js, Express, Docker, Docker Compose, GitHub Actions, AWS EC2, Nginx*

- Built and containerized a Node.js/Express uptime-monitoring service with Docker, checking service health every 60 seconds and sending Discord webhook alerts on downtime.
- Set up an end-to-end CI/CD pipeline with GitHub Actions — automated testing and image builds on every push, and automatic build, push to Docker Hub, and deployment to AWS EC2 on merges to main.
- Configured Nginx as an L7 reverse proxy/load balancer across multiple app replicas and documented a path to production-grade high availability (ALB, Auto Scaling, health checks).

Academic Projects

Full-Stack Cafe Management System

2026

Language and Tech Used: *Java, Jakarta EE, JDBC, MySQL, JSP/JSTL, Eclipse*

- Built a full-stack cafe management system with Java and Jakarta EE, integrating MySQL via JDBC.
- Implemented transaction management for data integrity and batch processing for bulk database updates.
- Followed MVC architecture with dynamic JSP/JSTL web interfaces for maintainability and scalability.

Network Traffic Analysis — CICIDS 2017

2026

Language and Tech Used: *Python, NumPy, Pandas, Matplotlib, Seaborn, SciPy*

- Analyzed 488,115 network flow records, performing data cleaning, EDA, and statistical hypothesis testing.
- Conducted a two-sample Z-test concluding Flow Duration alone is not a significant feature for intrusion detection.
- Visualized feature distributions and traffic patterns across Normal and Infiltration Attack classes.

Transport Management System — Database Design (SajiloYatra)

2025

Language and Tech Used: *SQL, Oracle*

- Designed ER diagrams and translated them into relational schemas normalized to 3NF.
- Implemented full CRUD operations and complex SQL queries to validate the design.

Art Gallery Management Application

2024

Language and Tech Used: *Java, Swing, JWT*

- Designed a desktop Art Gallery Management System with a responsive Swing GUI.
- Implemented JWT-based authentication, role-based permissions, visitor session management, and billing records.

Inventory Management System — SpeedzWear

2024

Language and Tech Used: *Python*

- Built a file-driven inventory management app with modular design, input validation, and exception handling.
- Implemented search, stock administration, and a menu-driven interface for a wholesale sneaker reseller.

CERTIFICATIONS

- [AWS Academy Graduate — Cloud Foundations](#)
- [AWS Academy Graduate — Data Engineering](#)
- [AWS Academy Graduate — ML for NLP](#)
- [AWS Academy Graduate — ML Foundations](#)